

Humberto Bernal, Ph.D.

Associate Professor | Industrial Organization & Regulatory Economics

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Professional Profile

Applied Economist and University Professor with a Ph.D. in Economics from Universidad de los Andes and over 20 years of research and teaching experience across academic, multilateral, and governmental institutions. Specializes in empirical **Industrial Organization**, **Regulatory Economics**, and **Game-Theoretic Mechanism Design**, with particular focus on natural monopolies, infrastructure concessions, market competition, and firm agglomeration. Combines rigorous theoretical modeling with quasi-experimental econometrics and computational methods (Python, R, Stata, MATLAB, SQL). Proven track record leading empirical evaluations for the Inter-American Development Bank (IDB) and public agencies, authoring peer-reviewed research published in leading international journals (Oxford Academic, Taylor & Francis), and mentoring undergraduate and graduate researchers with teaching evaluations consistently above 4.5/5.0.

Research Fields & Core Competencies

- **Industrial Organization & Competition Policy:** Empirical analysis of firm clustering, external economies of scale, market structure, barrier-to-entry dynamics, and strategic firm interaction.
- **Regulatory Economics & Infrastructure:** Economic regulation of natural monopolies, concession contract design (4G highway concessions), public-private partnerships, transportation networks, and natural resource extraction shocks (crude oil and gas).
- **Applied Microeconomic Theory & Mechanism Design:** Dynamic screening models under asymmetric information, incentive-compatible mechanisms, agent rationality under uncertainty, and computational interactive simulation (Streamlit).
- **Econometrics & Causal Inference:** Quasi-experimental identification (Difference-in-Differences, Instrumental Variables, Regression Discontinuity, Fixed Effects), microeconometrics, dynamic panel data, and Leontief input-output multiplier analysis.

Academic and Professional Experience

Universidad Colegio Mayor de Cundinamarca	2019 – Present
<i>Professor Coordinator & University Professor</i>	

- Coordinate the Economics faculty team, leading curricula modernization in microeconomics, econometrics, and quantitative analysis aligned with international academic standards.
- Direct empirical research on spatial agglomeration and transport infrastructure in Colombia, developing quasi-experimental econometric models to evaluate regional economic impacts.
- Formulate dynamic mechanism design models to identify rational agent types under uncertainty, deploying open-source computational simulations on Streamlit ([Interactive App](#)).
- Teach upper-level undergraduate courses in Microeconomics, Macroeconomics, and Economic Theory, consistently achieving student evaluation ratings above 4.5/5.0.

G&D Gerencia y Dirección de Proyectos S.A.S.	June 2022 – Dec 2022
<i>Project Manager</i>	

- Directed the implementation of the National Strategy for Socio-Occupational Orientation for Colombia's Ministry of National Education, managing project milestones, indicators, and cross-functional teams.
- Structured empirical evaluation frameworks and regional labor market diagnostics across national datasets.
- Synthesized multi-source empirical findings into comprehensive policy reports delivered to national government stakeholders.

- Directed the INSOR Data Desk, architecting statistical pipelines and econometric models on national microdata (GEIH, DANE) with hundreds of thousands of records to evaluate social and economic policy effectiveness.
- Formulated econometric wage-gap and labor-market access models (decomposition methods, logistic regression), identifying structural barriers for marginalized populations.
- Translated empirical findings into high-impact public policy briefs, culminating in a lead Q1 publication in *The Journal of Deaf Studies and Deaf Education* (Oxford University Press).

Inter-American Development Bank (IDB) / CNC / Multilateral Research Projects 2008 – 2018

External Researcher / University Professor

- Conducted socio-economic evaluations and regulatory impact analyses of multi-million-dollar highway infrastructure concessions in Colombia and Nicaragua for the IDB and Centro Nacional de Consultoría.
- Modeled regional commodity market dynamics, isolating econometric shocks of crude oil and natural gas production on municipal GDP, employment, and fiscal revenues for Universidad de los Andes.
- Taught undergraduate and graduate courses in Microeconomics and International Economics at Universidad Externado de Colombia, lecturing to cohorts of over 40 students per semester.
- Authored technical reports and economic advantage studies for international entities including Euromonitor International, PROCOLOMBIA, Fedesarrollo, and Universidad de los Andes.

Education

Ph.D. in Economics, Universidad de los Andes, Bogotá, Colombia, 2015.

M.Sc. in Economics, Universidad de los Andes, Bogotá, Colombia, 2005.

B.S. in Economics, Universidad Externado de Colombia, Bogotá, Colombia, 2003.

Academic Honors: Best Student Scholarship (2000, 2003).

Selected Publications & Working Papers

- **Bernal, H. (2026)**. Identifying Rational Types in Unknown Environments under an Indirect Dynamic Mechanism. *Working Paper* (under review at *International Journal of Game Theory*). Interactive model: [Streamlit App](#).
- **Bernal, H. (2025)**. Empowering deaf rights: enhancing access to Colombia's labor market. *The Journal of Deaf Studies and Deaf Education* (Oxford Academic), 30(4), 428–445. [doi:10.1093/jdsade/enaf034](#)
- **Bernal, H. (2022)**. Intra-industry agglomeration and the external economies of scale model: empirical evidence from Colombia. *Spatial Economic Analysis*, 17(3), 321–338. [doi:10.1080/17421772.2021.2009905](#)
- **Bernal, H. (2020)**. Demand and Supply in the Cocaine Market: An Empirical Study. *Journal of Globalization and Development*, 11(1). [doi:10.1515/jgd-2018-0047](#)
- **Bernal, H. (2019)**. Concesiones viales de cuarta generación (4G): metodología para cuantificar su impacto económico y social. *Revista de Economía & Región*, 13(2). [doi:10.32397/er.vol13.n2.1](#)

Languages & Professional Credentials

- **Languages**: Spanish (Native); English (Professional proficiency, IELTS).
- **Continuous Professional Development**: Computational Methods for Mechanism Design, LLM Benchmarking, and Applied AI Workflows (2024–2026); Advanced English, Language Link (2007).